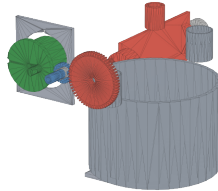


OPTION F

A Beginner's Guide to the ~1/3-Scale, 28BYJ-48 Pan/Tilt Redesign



This guide explains, in plain language, the Option F pan/tilt mechanism -- a smaller rebuild of the original **ASSEMBLY_pan_tilt** reference design, scaled to about 1/3 size and built around two real, off-the-shelf 28BYJ-48 stepper motors (one for pan, one for tilt) instead of the original's generic placeholder motor. It has the same 16 unique part designs and the same two-axis mechanism as the original: it swivels left-right (**pan**) and has a second, separate motor and gear set that produces a tipping motion (**tilt**), with a small clamp for a laser module on top. No prior mechanical design or CAD experience is assumed -- every technical word is explained the first time it is used.

Part types / total pieces	16 types / 24 pieces (matches the original's own count exactly)
Overall assembled size (measured)	146 x 114 x 112 mm (X x Y x Z)
Size vs. the original (245 x 298 x 270mm)	bigger than a literal 1/3 scale (~82 x 99 x 90mm) -- see Chapter 3 for exactly why, and why that was accepted rather than fixed
Pan / tilt gear reduction	3 : 1 (each axis, both verified)
Motor	Real 28BYJ-48 stepper (x2) -- the original used an unbranded placeholder can

How this guide is organised: Chapter 1 introduces every physical part with a picture and a number, plus an exploded view showing how they all separate. Chapters 2 and 3 refer back to those same part numbers throughout, so you can always flip back to Chapter 1 if you forget what "Part 6" is.

Chapter 1 - Parts of the Design

Every physical piece of this mechanism is listed below with a number, a plain-language description, and how it's meant to be made. Parts labelled **3D PRINTED** are meant to be printed on an ordinary home FDM printer; parts labelled **BOUGHT** are real, off-the-shelf items you would order rather than make -- the motors, the ball bearings, and the small motor-shaft bushings.

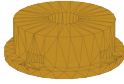
#	Part name	Type	Qty
1	Main Housing (+2 tilt-bearing towers)	3D PRINTED	1
2	Pan-Shaft Support Bearing (F625ZZ)	BOUGHT	1
3	Pan Shaft	3D PRINTED	1
4	Pan Motor Mounting Plate	3D PRINTED	1
5	28BYJ-48 Stepper Motor	BOUGHT	2
6	Motor Shaft Support Bushing	BOUGHT	2
7	Motor Shaft Coupler	3D PRINTED	2
8	Standoff Spacer	3D PRINTED	4
9	Tilt Motor Mounting Bracket	3D PRINTED	1
10	Tilt-Shaft Wall Bearing (625ZZ)	BOUGHT	2
11	Tilt Shaft - drive side	3D PRINTED	1
12	Tilt Gear, 60 tooth	3D PRINTED	1
13	Tilt Shaft - hub side	3D PRINTED	1
14	Payload Fork ("Cam Holder") + laser mount	3D PRINTED	1
15	Pinion Gear, 20 tooth	3D PRINTED	2
16	Pan Gear, 60 tooth	3D PRINTED	1



1. Main Housing (+2 tilt-bearing towers)

3D PRINTED Qty: 1

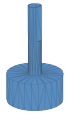
The fixed outer drum the whole mechanism is built inside and on top of. Widened beyond a literal 1/3 scale (73mm to 87mm across) so the real 28mm motor can and its 24mm gear-mesh distance actually fit inside - a direct, necessary consequence of using a real motor rather than a shrink-to-fit placeholder. Two integral posts on top carry the tilt axis's bearings.



2. Pan-Shaft Support Bearing (F625ZZ)

BOUGHT Qty: 1

A real, off-the-shelf small flanged ball bearing (5mm bore, 16mm body, 19mm mounting flange) - the closest real part to the original's much bigger four-bolt flange bearing at this scale. Its flange bolts to the housing; its bore lets the Pan Shaft spin freely.



3. Pan Shaft

3D PRINTED Qty: 1

The shaft running up through the housing's middle. Its lower stem rides in the pan bearing; its upper hub carries the big Pan Gear and forms the rotating platform everything pan-side sits on.



4. Pan Motor Mounting Plate

3D PRINTED Qty: 1

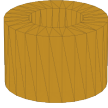
A bracket that holds the pan motor inside the housing. Its two mounting holes are deliberately shifted about 8mm off the motor's own centreline - the real 28BYJ-48's output shaft is not in the middle of the can, so the holes are moved to bring the shaft back onto the pan axis.



5. 28BYJ-48 Stepper Motor

BOUGHT Qty: 2

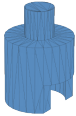
A real, off-the-shelf 28BYJ-48 stepper motor - one drives pan, one drives tilt. Modelled at its true size (28mm can, 35mm mounting-tab spacing, 8mm off-centre shaft); it is bought hardware, so unlike the printed housing around it, it is not shrunk to any scale.



6. Motor Shaft Support Bushing

BOUGHT Qty: 2

A small sleeve bushing steadying each motor's own shaft near its pinion gear. The original design's equivalent (a tiny ball bearing) doesn't exist as a real, buyable part once shrunk to this scale, so a plain bushing takes its place.



7. Motor Shaft Coupler

3D PRINTED Qty: 2

A small stepped coupler: one end clamps onto the real motor's D-shaped shaft, the other end carries the small pinion gear that drives the big gear.



8. Standoff Spacer

3D PRINTED Qty: 4

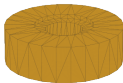
Short printed sleeves keeping hardware positioned correctly along each motor's shaft/coupler assembly - two per motor, matching the original design's four-spacer count exactly.



9. Tilt Motor Mounting Bracket

3D PRINTED Qty: 1

The bracket that holds the tilt motor. Like the pan motor's plate, its mounting holes are shifted to line up with the real motor's off-centre shaft rather than an idealised centred one.



10. Tilt-Shaft Wall Bearing (625ZZ)

BOUGHT Qty: 2

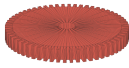
A real, off-the-shelf 625ZZ ball bearing (5mm bore, 16mm OD) - this one maps almost exactly onto the original design's own bearing size once scaled down, a clean substitution. One seats in each of the housing's two tilt-axis posts.



11. Tilt Shaft - drive side

3D PRINTED Qty: 1

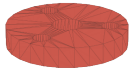
One half of the tilt axle. Bolted rigidly to the Payload Fork with 3 screws (matching the original design's own bolt pattern, verified from its geometry) and carries a small spigot that the tilt bearing mounts on.



12. Tilt Gear, 60 tooth

3D PRINTED Qty: 1

The big gear that produces the tilt motion, meshing with the tilt motor's small pinion gear.



13. Tilt Shaft - hub side

3D PRINTED Qty: 1

The other half of the tilt axle, directly opposite Part 11 on the same axis, also bolted rigidly to the Payload Fork.



14. Payload Fork ("Cam Holder") + laser mount

3D PRINTED Qty: 1

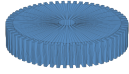
The fork spanning across the housing's two tilt-axis posts, bolted rigidly (not pivoting) to both tilt shaft halves - confirmed from the original file's own geometry, where the same bolt pattern lines up on both parts. A clamp on top holds a small cylindrical laser module.



15. Pinion Gear, 20 tooth

3D PRINTED Qty: 2

A small gear on each motor's coupler shaft. One drives the pan gear, the other the tilt gear - the small gear turns 3 times for every 1 turn of the big gear it drives (see Chapter 3).



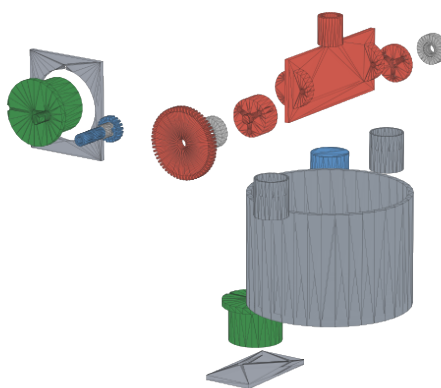
16. Pan Gear, 60 tooth

3D PRINTED Qty: 1

The big gear that produces the pan motion, mounted on the Pan Shaft's hub and meshing with the pan motor's pinion.

Exploded View

The picture below pulls all 24 pieces apart along the lines they actually sit on in the real assembly, so you can see how each one relates to its neighbours without anything hidden behind another part. Nothing about the parts themselves changes -- this is the same model as the hero picture on the title page, just pulled apart for clarity.

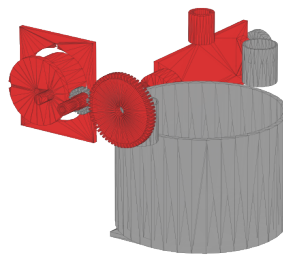


Chapter 2 - Movement

This mechanism moves in two intended directions, called axes:

Pan - swivels the top of the mechanism left and right, like shaking your head 'no'. The Pan Shaft (Part 3) and Pan Gear (Part 16) spin about a vertical axis running up through the housing.

Tilt - a second, separate motor and gear pair (Parts 9-15, tilt copies) tips the Payload Fork and its laser mount up and down, like nodding your head 'yes', about a horizontal axis carried in the housing's own twin posts.



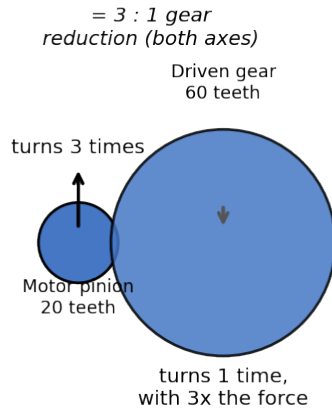
Grey = fixed, never moves Blue = moves with PAN Red = moves with TILT

#	Part	How it moves
1	Main Housing (+2 tilt-bearing towers)	Fixed - never moves. Everything else is built on or inside it (including both tilt-axis posts).
2	Pan-Shaft Support Bearing (F625ZZ)	Outer flange fixed in the housing; inner race spins with the Pan Shaft.
3	Pan Shaft	Spins about the vertical axis - this spin IS the pan motion.
4	Pan Motor Mounting Plate	Bolted fixed to the housing; never moves.
5	28BYJ-48 Stepper Motor	One copy is fixed (bolted to the pan motor plate); the other is also fixed (bolted to the tilt motor bracket, which is itself fixed to the housing in this model - see the engineering note in Chapter 3).
6	Motor Shaft Support Bushing	Same split as the motor bodies - both copies are fixed in this model.
7	Motor Shaft Coupler	The pan-side copy spins in place to drive the pan gear; the tilt-side copy spins in place to drive the tilt gear.
8	Standoff Spacer	Fixed; two hold the pan motor's shaft train, two hold the tilt motor's.
9	Tilt Motor Mounting Bracket	Bolted fixed to the housing; never moves.
10	Tilt-Shaft Wall Bearing (625ZZ)	Outer race fixed in the housing's tilt-axis post; inner race spins with the tilt shaft.

11	Tilt Shaft - drive side	Spins about the horizontal tilt axis - this rotation IS the tilt motion.
12	Tilt Gear, 60 tooth	Spins about the tilt axis with Part 11 - this rotation IS the tilt motion.
13	Tilt Shaft - hub side	Spins about the tilt axis, opposite Part 11 on the same line.
14	Payload Fork ("Cam Holder") + laser mount	Rigidly bolted to Parts 11 and 13; pivots with them - this whole fork IS the tilt motion output, carrying the laser mount with it.
15	Pinion Gear, 20 tooth	The pan-side copy spins in place to drive the pan gear; the tilt-side copy spins in place to drive the tilt gear.
16	Pan Gear, 60 tooth	Spins about the vertical pan axis with the Pan Shaft - this rotation IS the pan motion.

Chapter 3 - How the Motion Is Actually Produced

Both axes use exactly the same trick, just aimed in different directions. A small 20-tooth gear (the pinion, Part 15) is fixed to each motor's own coupler shaft, and it meshes with a much bigger 60-tooth gear (Part 16 for pan, Part 12 for tilt). A gear is a toothed wheel that locks its teeth into a matching wheel so the two always turn together at a fixed ratio. This is called a **gear reduction**:



Because the big gear has three times as many teeth as the small one, the small gear must spin all the way round three times to make the big gear turn once. That trade means the gear pair also multiplies turning force (**torque** -- a twisting force, like the effort needed to turn a stiff doorknob) by 3x. The original design's gear teeth used a 1.125mm module (a standard size unit for gear teeth); scaled to a literal 1/3 that would be an unprintable 0.375mm, so this redesign uses a 0.6mm module instead -- the tooth counts (20 and 60) stay exactly the same, so the 3:1 ratio is unchanged, but the gears end up sitting 24.0mm apart instead of a literal-thirds 15mm, a deliberate, accepted trade for printability.

Engineering note - why this model is bigger than a literal 1/3 scale: the overall assembled size is 146 x 114 x 112mm, well past the ~82 x 99 x 90mm a literal 1/3 scale would give -- most of that comes from one traceable cause. Shaft 2 and Baring 2 Shaft (Parts 11 and 13) each carry a hub sized to comfortably fit the real 16mm-OD tilt bearing. That hub, the fixed bearing post around it, and the Tilt Gear (Part 12) all have to clear the rigid Payload Fork's own solid arm, one after another, on the same side -- and each clearance step adds real millimetres. This was found and measured during assembly, reviewed, and explicitly **accepted as final** rather than corrected: shrinking that hub would shrink the size back down, and is recorded as a known follow-up if a tighter fit is wanted later.

Engineering note - the payload fork's joint, and the pan/tilt independence question: the original reference file's own "Cam Holder" part was measured directly from its geometry and found to be bolted rigidly to its tilt shaft with 3 matching screw holes on each side, not pivoting on a separate joint -- this redesign's Payload Fork (Part 14) copies that same rigid, 3-bolt connection deliberately, per an explicit decision made before any of this model was built. Separately: in the original design, the whole tilt stage is carried around bodily whenever the pan motor turns. In this model, the two tilt-axis posts and the tilt motor bracket (Part 9) are built into the fixed housing itself rather than onto the rotating Pan Shaft's hub -- so, as currently modelled, panning turns the Pan Shaft and Pan Gear in place but does not carry the tilt stage around with it. This is a simplification from the assembly pass, flagged here honestly rather than left undocumented, and is a candidate for a future revision if full pan-carries-tilt behaviour is wanted.

Chapter 3 (continued) - Assembly

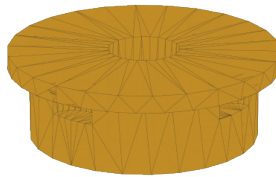
Build the mechanism outward from the housing. Two kinds of joints come up throughout:

Fixed / rigid joint - bolted, keyed, or press-fit so the two parts never move relative to each other again.

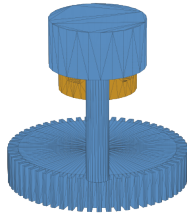
Moving joint - a bearing or a shaft spinning inside a hole, deliberately free to rotate.

Step 1 - Fit the pan bearing and towers

Start with the Main Housing (Part 1), which already has the two tilt-axis posts built in. Press the pan bearing (Part 2) into the register on top - moving joint, its flange is fixed in the housing, its bore is what the Pan Shaft will spin inside.

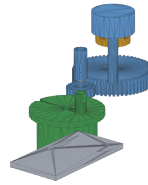


Step 2 - Fit the pan shaft and pan gear



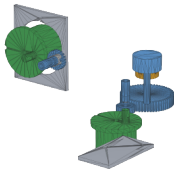
Slide the Pan Shaft (Part 3) down through the pan bearing from above - moving joint, it should spin freely once through. The Pan Gear (Part 16) is carried on the shaft's own hub - fixed joint, gear and shaft always turn together.

Step 3 - Add the pan motor



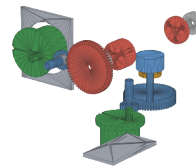
Fix the Pan Motor Mounting Plate (Part 4) inside the housing - fixed joint, it never moves. Fit the motor body, its support bushing, coupler, and 20-tooth pinion (Parts 5/6/7/15, pan copies) and check the pinion meshes with the pan gear.

Step 4 - Add the tilt motor bracket



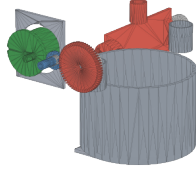
Fix the Tilt Motor Mounting Bracket (Part 9) to the housing near the tilt-axis posts - fixed joint. Fit its own motor body, bushing, coupler, and pinion (Parts 5/6/7/15, tilt copies) the same way as Step 3.

Step 5 - Fit the tilt bearings and tilt shafts



Press the two Tilt-Shaft Wall Bearings (Part 10) into the housing's twin posts - moving joints. Fit the drive-side and hub-side tilt shaft halves (Parts 11 and 13) into them, with the Tilt Gear (Part 12) keyed to the drive side and meshing with the tilt pinion.

Step 6 - Fit the Payload Fork and laser mount



Bolt the Payload Fork (Part 14) onto both tilt shaft halves with 3 screws on each side - fixed joint, confirmed from the original design's own geometry as a rigid connection, not a separate pivot. Clip the small cylindrical laser module into the clamp on top.

This is a beginner-oriented, plain-language explanation of the Option F pan/tilt CAD model (OPTION_F.step), built as a ~1/3-scale redesign of ASSEMBLY_pan_tilt.STEP around two real, off-the-shelf 28BYJ-48 stepper motors, with a small clamp added for a laser module payload. Every measurement above was taken directly from the actual, already-exported model's geometry, not guessed. This is a design-intent / layout-study model (bought components and gear teeth are simplified, correctly-sized stand-ins, not release-ready manufacturing geometry) -- real-world fits, tolerances, and print settings should be checked before manufacturing anything from it.